

Accuracy of Adaptive Order WENO Reconstructions for Solving Conservation Laws

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Conservation Law

$$u_t + f(u)_x = 0, \quad t > 0, \quad x \in \mathbb{R}, \quad u \in \mathbb{R}^d, \quad d \geq 1.$$

Question

Do WENO reconstructions drop to lower order when there is a shock or contact discontinuity?

Polynomial Approximation on Stencils

$$S_j^r(i) = \{I_{i+j-\lfloor \frac{r}{2} \rfloor}, \dots, I_i, \dots, I_{i+j+\lfloor \frac{r-1}{2} \rfloor}\}$$

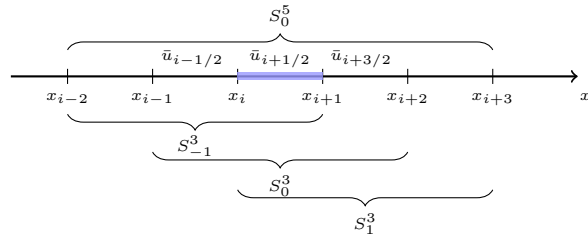
Obtain the r th order stencil polynomial $P_j^r \in \mathcal{P}^{r-1}$ by imposing the interpolation conditions

$$\frac{1}{\Delta x_k} \int_{I_k} P_j^r(x) dx = \bar{u}_{k+1/2}, \quad \forall I_k \in S_j^r.$$

Smoothness Indicators (Jiang and Shu, 1996)

$$\sigma_j^r = \sum_{\ell=1}^{r-1} \int_{I_i} (\Delta x_i)^{2\ell-1} \left(\frac{d^\ell P_j^r(x)}{dx^\ell} \right)^2 dx.$$

Two Level WENO-AO(5,3)



$$R_3^{5,3}(x) = \frac{\tilde{\alpha}_0^5}{\alpha_0^5} \left[P_0^5(x) - \sum_{j=-1}^1 \alpha_j^3 P_j^3(x) \right] + \sum_{j=-1}^1 \tilde{\alpha}_j^3 P_j^3(x),$$

where $\alpha_0^5 + \sum_j \alpha_j^3 = 1$.

WENO-JS weights

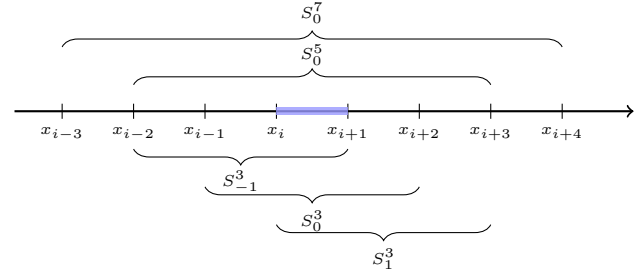
$$\hat{\alpha}_{r_j}^r = \frac{\alpha_{r_j}^r}{(\sigma_{r_j}^r + \epsilon_h)^\eta}, \quad \tilde{\alpha}_{r_j}^r = \frac{\alpha_{r_j}^r}{\sum_{s,k} (\sigma_{s_k}^r + \epsilon_h)^\eta}, \quad r, s = 3, 5.$$

$$\text{WENO-Z weights} \quad \tau = \frac{1}{3} \sum_j |\sigma_0^5 - \sigma_j^3|,$$

$$\hat{\alpha}_{r_j}^r = \alpha_{r_j}^r \left(1 + \left(\frac{\tau}{\sigma_{r_j}^r + \epsilon_h} \right)^\eta \right), \quad \tilde{\alpha}_{r_j}^r = \frac{\alpha_{r_j}^r}{\sum_{s,k} (\sigma_{s_k}^r + \epsilon_h)^\eta}.$$

Multilevel WENO-AO₃(7,5,3)

(Balsara, Garain and Shu, 2016)



$$R_3^{7,5,3}(x) = \frac{\tilde{\gamma}^{7,3}}{\gamma^{7,3}} [R_3^{7,3}(x) - \gamma^{5,3} R_3^{5,3}(x)] + \tilde{\gamma}^{5,3} R_3^{5,3}(x),$$

where $\gamma^{7,3} + \gamma^{5,3} = 1$. (WENO-Z: $\tau = |\sigma_0^7 - \sigma_0^5|$.)

New Multilevel WENO-AO(7,5,3)

$$R^{7,5,3}(x) = \frac{\tilde{\alpha}_0^7}{\alpha_0^7} \left[P_0^7(x) - \sum_{j=-1}^1 \alpha_j^3 R^{5,3}(x) \right] + \sum_{j=-1}^1 \tilde{\alpha}_j^3 R^{5,3}(x),$$

where $\alpha_0^7 + \sum_j \alpha_j^3 = 1$. (WENO-Z: $\tau = \frac{1}{3} \sum_j |\sigma_0^7 - \sigma_j^3|$.)

Use P_0^7 instead of $R_3^{7,3}$ in the reconstruction.

Theorem: smooth case

WENO-JS weights

(+ sign: $\epsilon_h = \epsilon$, - sign: $\epsilon_h = K \Delta x_i^2, K > 0$.)

WENO-AO(r, s) achieves accuracy $\mathcal{O}(h^r)$ if $r - s \leq s \pm 1$.

WENO-AO_s($r_\ell, \dots, r_1, s$) achieves optimal convergence

$\mathcal{O}(r_\ell)$ if $r_\ell - s \leq s \pm 1$.

WENO-AO($r_\ell, \dots, r_0$) achieves optimal convergence if

$r_k - r_{k-1} \leq r_0 \pm 1, \forall 1 \leq k \leq \ell$.

WENO-Z weights

All of the reconstructions can achieve the optimal order by carefully choosing η .

Theorem: discontinuous case

Assume $\sigma \rightarrow 0$ as the grid is refined.

Take $K > 0$, $\epsilon_h = K h^2$, conditions on η .

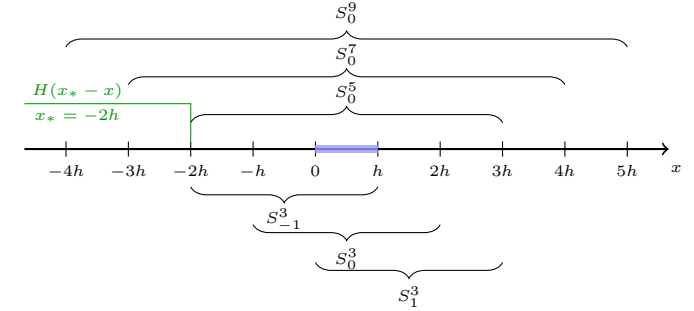
WENO-AO(r, s) drops to order s if $\eta \geq s/2$.

WENO-AO_s($r_\ell, \dots, r_1, s$) drops to base order s when u has a discontinuity on $r_{\ell-1}$.

WENO-AO($r_\ell, \dots, r_0$) drops to the accuracy of the largest stencil where u is smooth.

Example: WENO-AO_{*}(9,7,5,3)

$$u(x) = x^3 + \sin(x) + H(x_* - x).$$



Error & convergence rate at $x = 0$, $\epsilon_h = h_n^2$, $h_n = 0.1/2^n$.

WENO-JS weights

n	$x_* = -4h$		$x_* = -3h$		$x_* = -2h$	
	error	order	error	order	error	order
WENO-AO ₃ (9,7,5,3)						
6	2.91E-15	5.00	2.91E-15	5.00	4.25E-12	3.04
7	9.09E-17	5.00	9.10E-17	5.00	5.27E-13	3.01
8	2.84E-18	5.00	2.84E-18	5.00	6.58E-14	3.00
WENO-AO(9,7,5,3)						
6	3.57E-27	9.00	2.89E-21	7.00	3.07E-15	5.00
7	6.97E-30	9.00	2.26E-23	7.00	9.58E-17	5.00
8	1.36E-32	9.00	1.76E-25	7.00	2.99E-18	5.00

WENO-Z weights

n	$x_* = -4h$		$x_* = -3h$		$x_* = -2h$	
	error	order	error	order	error	order
WENO-AO ₃ (9,7,5,3)						
6	3.35E-26	9.00	2.38E-19	7.00	8.64E-12	3.04
7	6.54E-29	9.00	1.86E-21	7.00	1.07E-12	3.01
8	1.28E-31	9.00	1.45E-23	7.00	1.34E-13	3.00
WENO-AO(9,7,5,3)						
6	4.41E-29	9.00	7.79E-23	6.94	1.61E-16	5.05
7	8.60E-32	9.00	6.21E-25	6.97	4.93E-18	5.02
8	1.68E-34	9.00	4.91E-27	6.98	1.53E-19	5.01

Summary and Conclusions

The new reconstruction:

1. Achieves the optimal order of accuracy for a wider choice of levels;
2. Drops to the highest possible order of accuracy near the discontinuity, rather than the base level;
3. All reconstructions can fail on carefully constructed meshes ($\sigma \rightarrow 0$).